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- Polyacetal (POM) 101: Material Properties and Applications

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Complete Guide on Polyoxymethylene (POM) or Polyacetal Plastic

Polyacetal or Polyoxymethylene has a semi-crystalline nature. It is a high-performance thermoplastic polymer. Known for its exceptional mechanical properties and versatile applications. In this guide, we will help you make informed decisions when selecting materials. Get more insights on how to optimize the use of POM in your projects.

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[Overview](#)

What is a Polyacetal?

What is a Polyacetal?

Polyacetal is a formaldehyde-based, semi-crystalline engineering thermoplastic. It is commonly called acetal or polyoxymethylene (POM). It contains the functional group of a carbon bonded to two -OR groups. It has a

molecular structure of $(\text{CH}_2\text{O})_n$.

Molecular Structure of Polyoxymethylene (Chemical Formula: $(\text{CH}_2\text{O})_n$)

POM is 100% recyclable. It is also known as polyformaldehyde, polymethylene glycol, and polyoxymethylene glycol.

How was polyoxymethylene developed over the years?

How was polyoxymethylene developed over the years?

The key milestones achieved in the development of POM are:

1. **Discovery of Formaldehyde Polymers:** First synthesized and studied in the early 20th century. These polymers include acetal polymers. Researchers discovered that formaldehyde could polymerize to form materials with desirable properties.
2. **Development of POM:** In 1920, Hermann Staudinger, a German chemist discovered Polyoxymethylene. He conducted extensive research on polymers and the concept of macromolecules. His work laid the foundation for the development of polyacetal polymers. Later he received the Nobel Prize in Chemistry in 1953.
3. **Commercialization in 1956:** DuPont became the first company to produce a POM homopolymer. They introduced a method based on formaldehyde polymerization using a coordination catalyst.
4. **Commercialization in 1962:** Celanese became the first company to produce POM copolymer. They employed an acid-catalyzed process to produce this copolymer.

Over the decades new developments in polyoxymethylene (POM) have taken place. These advancements are with regard to polymer chemistry, industrial manufacturing techniques, and properties. View all [POM commercial grades and suppliers](#) »

How to produce acetal resins?

How to produce acetal resins?

Acetal resins are produced by the polymerization of purified formaldehyde $[\text{CH}_2\text{O}]$. However, different manufacturing processes are used to produce the homopolymer and copolymer versions of POM. In alkaline environments, copolymers are more stable than the homopolymers. Yet homopolymers provide better

mechanical properties than copolymers.

POM is available in different forms. Homopolymer resins include:

- DowDuPont - [Derlin®](#)
- Asahi Kasei Plastics - [Tenac®](#)

And, popular copolymer resins are available under the following trade names:

- Celanese - [Celcon®](#)
- Polyplastics - [Duracon®](#)
- BASF - [Ultraform®](#)

How to compare between POM homopolymer or copolymer?

How to compare between POM homopolymer or copolymer?

Acetal homopolymer is produced from anhydrous, monomeric formaldehyde which is polymerized by anionic catalysis in an organic liquid reaction medium. The resulting polymer is stabilized by the reaction to acetic anhydride.

While, the **copolymer of POM** requires the conversion of formaldehyde into trioxane using acid catalysis and cationic polymerization. The reaction is followed by purification of the trioxane by distillation or extraction to remove water and other active impurities containing hydrogen.

Acetal Copolymer	Acetal Homopolymer
• Easier to process / wider processing window • Superior long-term performance (creep resistance, fatigue, endurance, strength retention) • Less gassing and odor • Heavy metal free colors, i.e. cadmium and lead (safer for workers / environment) • Better maintenance of color • Under ultraviolet light exposure • Faster molding cycles • Less mold deposits • Stable in alkaline environments	• Available in several viscosity ranges • Greater degree of regularity in their structure • Higher tensile strength • Unfilled homopolymer is stiffer and stronger • Moderate toughness under repeated impact • Allows thinner and lighter part design • Shorter molding cycles • Potential for cost reductions • Provide better mechanical properties
POM copolymer grades »	POM homopolymer grades »

What are the key properties of acetal resins?

What are the key properties of acetal resins?

Polyoxymethylene resins demonstrate well-balanced properties ranging from mechanical to physical and flammability performance. The key benefits of POM resins include:

- Excellent mechanical properties over a temperature range upto 140°C, down to -40°C
 - High tensile strength, rigidity and [toughness](#) (short-term)
 - Low tendency to creep (as compared to nylon) and fatigue (long-term). Not susceptible to environmental stress cracking
- High degree of crystallinity and excellent dimensional stability
- [High gloss surfaces](#)
- Excellent wear resistance
- Good resistance to organic solvents and chemicals (except phenols) at room temperature
- Low smoke emission
- Low coefficient of friction
- [Low moisture absorption](#)

What are the different POM grades available?

What are the different POM grades available?

POM grades are often produced with various degrees of polymerization. This results in different properties to meet demanding applications. The different grade options of POM resins are:

1. **Standard/Unreinforced Grades**
2. **Reinforced Grades:** These grades show high tensile strength or rigidity. These properties depend on the type and amount of polymer reinforcement. [Glass fibers](#), [carbon fibers](#) or glass spheres-reinforced POM grades are available.
3. **High-Impact/Toughened Grades:** Blending POM resins with rubber, TPU, and other polymers. This results in blends with higher impact strength.
4. **Grades with High Slip/Wear Properties:** Modification of POM with graphite, PTFE, and mineral fillers. These additives enhance abrasion resistance and slip properties.
5. **UV Stabilized Grades:** UV stabilizers and absorbers are often added to POM resins or blends to [improve UV stability](#). e.g., hindered-amine light stabilizers.
6. **Nanocomposites:** Additives, such as CNTs, POSS, ZnO, etc. are used to produce POM nanocomposites.
7. **Other Grades:**
 - a. Addition of powdered Al or bronze enhances electrical conductivity or [heat distortion point](#) of POM resins.
 - b. Fluorocarbons lead to good surface lubricity in polyacetal to prevent cracking.

What are the benefits of POM over metals or thermoplastics?

What are the benefits of POM over metals or thermoplastics?

Check out the benefits of acetal resins over metals and other thermoplastics below.

Benefits over Metals	Benefits over Thermoplastics
• Design flexibility • High strength-to-weight ratio • Color matching possibility • Inherent lubricity • Lower finished part cost • Opportunities for parts consolidation • Chemical/corrosion resistance	• Low coefficient of friction • Good creep resistance • Good toughness/impact resistance • Hard surface with good appearance • High strength and stiffness • Excellent dimensional stability • Excellent chemical resistance

Processing Techniques for POM

Processing Techniques for POM

Polyacetal resins are supplied in a granulated form. They can be molded into a desired shape by applying heat and pressure. They can be processed by injection molding, extrusion, compression molding, rotational casting, or blow molding. [Injection molding](#) and [extrusion](#) are the most commonly used methods for POM processing.

POM resins must be processed in the temperature range (190 – 230°C). They may require drying before forming because of its hygroscopic nature.

Processing Conditions for Injection Molding

- Melt temperature
 - Homopolymer resins: 180-230°C
 - Copolymer resins: 190-210°C
- Mold temperature: 50-150°C. Use higher mold temperatures for precision molding for reduced post-molding shrinkage.
- Injection pressure: 70-120 MPa
- Injection speed: Medium to high

Extrusion Processing Conditions

Extrusion is used to produce semi-furnished parts, such as sheets, rods, pipes, filaments, & profile sections. They

are further machined using traditional methods such as turning, milling, drilling, etc. to form finished parts.

- Melt temperature: 180-230°C
- Screw speed: 33-42
- Die temperature: 175-230°C

Lightly crosslinked grades are used to produce hollow molding by blow molding.

[Easily Avoid Disruptions in Your Injection Molding or Extrusion Production Line - Watch Free Video Today!](#)

3D Printing of Acetal Grades

- Acetal has found some in-roads into [3D printing](#) in some applications like fan blade, impeller, etc.
- Its high lubricity surface (with 3-5% on average and as high as 7-10%) makes it interesting for 3D printing especially for difficult to release parts.
- Also, acetal polymers have high strength which assures dimensional stability up to a [maximum continuous service temperature](#) of 80°C (180°F).

What are the limitations of acetal polymer?

What are the limitations of acetal polymer?

- Poor resistance to strong acids, bases and oxidizing agents.
- Burns easily without flame retardants due to high oxygen content
- Poor thermal stability without suitable stabilizer system
- Limited processing temperature range
- High mold shrinkage
- Poor resistance to UV radiation. Prolonged exposure lead to color change, embrittlement, and loss of strength
- Low surface energy and hence difficult to bond without surface treatment. [Overcome bonding problem related to low surface energy substrates »](#)

What are the applications of POM?

What are the applications of POM?

POM resins have high lubricity, good dimensional stability, and sliding properties. Due to these properties, POM produces high-precision parts for applications such as:

- Automotive
- Electrical & electronic
- Industrial
- Drug Delivery

The polymer serves as an alternative to metals. This is due to its low friction, wear, and excellent balance of mechanical and chemical properties. Know more about [POM applications](#) in detail.

Key Properties

Key Properties

Property	UNREINFORCED POM	IMPACT MODIFIED POM	LOW FRICTION POM	MINERAL FILLED POM
Chemical Resistance				
Acetone @ 100%, 20°C	Limited	Limited	Limited	Limited
Ammonium hydroxide @ 30%, 20°C	Satisfactory	Satisfactory	Satisfactory	Satisfactory
Ammonium hydroxide @ diluted, 20°C	Satisfactory	Satisfactory	Satisfactory	Satisfactory
Ammonium hydroxide @ diluted, 60°C	Satisfactory	Satisfactory	Satisfactory	Satisfactory
Aromatic hydrocarbons @ 20°C	Limited	Limited	Limited	Limited
Benzene @ 100%, 20°C	Satisfactory	Satisfactory	Satisfactory	Satisfactory
Ethyleneglycol (Ethane diol) @ 100%, 20°C	Limited	Limited	Limited	Limited
Ethyleneglycol (Ethane diol) @ 100%, 50°C	Limited	Limited	Limited	Limited
Gasoline	Satisfactory	Satisfactory	Satisfactory	Satisfactory
Glycerol @ 100%, 20°C	Satisfactory	Satisfactory	Satisfactory	Satisfactory
Grease @ 20°C	Satisfactory	Satisfactory	Satisfactory	Satisfactory
Hydrogen peroxide @ 30%, 60°C	Non Satisfactory	Non Satisfactory	Non Satisfactory	Non Satisfactory
Kerosene @ 20°C	Satisfactory	Satisfactory	Satisfactory	Satisfactory

Methanol @ 100%, 20°C	Satisfactory	Satisfactory	Satisfactory	Satisfactory
Methylethyl ketone @ 100%, 20°C	Limited	Limited	Limited	Limited
Mineral oil @ 20°C	Satisfactory	Satisfactory	Satisfactory	Satisfactory
Phenol @ 20°C	Non Satisfactory	Non Satisfactory	Non Satisfactory	Non Satisfactory
Silicone oil @ 20°C	Limited	Limited	Limited	Limited
Sodium hydroxide @ 10%, 20°C	Satisfactory	Satisfactory	Satisfactory	Satisfactory
Sodium hydroxide @ 10%, 60°C	Limited	Limited	Limited	Limited
Sodium hypochlorite @ 20%, 20°C	Non Satisfactory	Non Satisfactory	Non Satisfactory	Non Satisfactory
Strong acids @ concentrated, 20°C	Non Satisfactory	Non Satisfactory	Non Satisfactory	Non Satisfactory
Toluene @ 20°C	Satisfactory	Satisfactory	Satisfactory	Satisfactory
Toluene @ 60°C	Satisfactory	Satisfactory	Satisfactory	Satisfactory
Xylene @ 20°C	Satisfactory	Satisfactory	Satisfactory	Satisfactory
Butylacetate @ 100%, 20°C	Satisfactory	Satisfactory	Satisfactory	Satisfactory
Chlorinated solvents @ 20°C	Limited	Limited	Limited	Limited
Chloroform @ 20°C	Non Satisfactory	Non Satisfactory	Non Satisfactory	Non Satisfactory
Diocetylphthalate @ 100%, 20°C	Limited	Limited	Limited	Limited
Ethanol @ 96%, 20°C	Satisfactory	Satisfactory	Satisfactory	Satisfactory

Electrical

Arc Resistance, sec	200-220	120	126-183	-
Volume Resistivity x 10 ¹⁵ , Ohm.cm	14-15	15-16	15-16	-
Dielectric Constant	3.3-4.7	4-4.3	3-4	-
Dielectric Strength, kV/mm	13.8-20	19	16	-
Dissipation Factor x 10 ⁻⁴	50-110	50-250	20-90	-

Mechanical

Flexural Modulus, Gpa	2.8-3.7	1.4-2.3	2-3	4-5.5
Hardness Rockwell M	75-94	35-79	58-94	83-90
Hardness Shore D	80-95	80-92	80-94	92-95
Strength at Break (Tensile), MPa	60-70	45-60	50-70	50-75
Strength at Yield (Tensile), MPa	54-78	35-50	48-69	54-78
Toughness at Low Temperature, J/m	53-250	-	32-53	-
Toughness, J/m	60-120	90-250	25-60	50-65
Young's Modulus, GPa	2.8-3.7	1.4-2.3	1.8-3	4-5.5

Elongation at Break, %	15-75	60-200	10-70	5-55
Elongation at Yield, %	8-23	10-15	-	-

Physical

Gamma Radiation Resistance	Fair	Fair	Fair	Fair
Glass Transition Temperature, °C	-60--50	-	-	-
Shrinkage, %	1.8-2.5	1-2.5	1.8-3	1.5-2
Sterilization Resistance (Repeated)	Poor	Poor	Poor	Poor
UV Light Resistance	Poor	Poor	Poor	Poor
Water Absorption 24 hours, %	0.15-0.5	0.3-0.5	0.2-0.27	0.2-0.5
Density, g/cm ³	1.41-1.42	1.3-1.35	1.4-1.54	1.5-1.6

Service Temperature

HDT @0.46 Mpa (67 psi), °C	158-172	145-165	168-172	158-175
HDT @1.8 Mpa (264 psi), °C	110-136	64-90	118-136	100-140
Max Continuous Service Temperature, °C	80-105	80-100	80-105	80-105
Min Continuous Service Temperature, °C	-40	-50--40	-40	-
Ductile / Brittle Transition Temperature, °C	-40	-50--40	-40	-

Thermal

Fire Resistance (LOI), %	18	18	-	-
Flammability, UL94	HB	HB	HB	HB
Thermal Insulation, W/m.K	0.31-0.37	-	0.31	-
Coefficient of Linear Thermal Expansion x 10 ⁻⁵ , /°C	10-15	12-13	10-12	8-9

[Key Applications](#)

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David R Aug 3, 2022 ☐ ☐ ☐ ☐ ☐ ☐ ☐ ☐ ☐

Very good article

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